

Decrypting Symmetric Encryption Using John the Ripper or Hashcat

Jack Irvin

Step 1: Generating a Weakly Encrypted ~~DES~~ (AES) (MD5) File

1. Create a Secret Message `echo "The password to the vault is 5f4dcc3b." > secret_message.txt`

```
root@JackLinux:~# echo "The password to the vault is 5f4dcc3b." > secret_message.txt
```

2. Encrypt the File Using MD5 (Weak Algorithm) `openssl enc -md5 -salt -in secret_message.txt -out encrypted_md5.enc -pass pass:123456`

```
root@JackLinux:~# openssl enc -des-ede3 -salt -in secret_message.txt -out encrypted_des.enc -pass pass:123456
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
```

3. Verify the Encrypted File `ls -l encrypted_aes.enc`

```
root@JackLinux:~# ls -l encrypted_des.enc
-rw-r--r-- 1 root root 56 Sep 21 21:57 encrypted_des.enc
```

Step 2: Extracting the Encrypted Hash for Cracking

Option 1: Extracting the Hash for John the Ripper

1. Convert the DES-encrypted file into a crackable format: `openssl enc -d -md5 -in encrypted_md5.enc -k 123456 -P` This will output something like: `salt=F3E5B7A9 key=6F57A623AA5F3E4A 12`

```
root@JackLinux:~# openssl enc -d -aes-256-cbc -in encrypted_aes.enc -k 123456 -P
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
salt=00332A4763D3ED31
key=74A919DBB80E5EC49BA102C2DE23AED28FC03A0E92D82A7DA0F965EB41691290
iv =6A4F25842A80EE28AEE3FE3BABA00F60
```

2. Format the hash: John the Ripper requires the hash in aes-256-cbc format, which can be manually added to a file.

```
root@JackLinux:~# echo '$aes256$53616c7465645f5f00332a4763d3ed31$2236046f3072a2cad9c493d68325837eba66fba4619c93c873cb537fd3c6d' > aes_hash.txt
root@JackLinux:~#
```

Step 3: Cracking the DES-encrypted File with John the Ripper

1. Install John the Ripper (if not installed) `sudo apt install john`

```
root@JackLinux:~# sudo apt install john
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
john is already the newest version (1.9.0-2build1).
0 upgraded, 0 newly installed, 0 to remove and 156 not upgraded.
root@JackLinux:~#
```

2. Create a password list (if you want to use a dictionary attack): `echo "123456" > password_list.txt` `echo "password" >> password_list.txt` `echo "secret" >> password_list.txt`

```
root@JackLinux:~# echo "123456" > password_list.txt echo "password" >> password_list.txt echo "secret" >> password_list.txt
root@JackLinux:~# ls
aes_hash.txt          decrypted_file.txt    des_bruteforce.py    encrypted_blowfish.enc  openssl12john.py
ciphertext_hex.txt    decrypted_message.txt des_hash.txt          encrypted_des.enc       password_list.txt
decrypted_blowfish.txt des3_hash.txt         encrypted_aes.enc     encrypted_message.enc   secret_message.txt
```

3. Run John the Ripper on the AES hash `john --wordlist=password_list.txt aes_hash.txt`

No matter what I did, I couldn't get John the ripper to run and crack a hash. At first it wanted DES which I couldn't use because it was insecure, then when I changed it to aes it couldn't crack the password. I finished the lab by using Hashcat instead because the lab presented the option of using John the Ripper or Hashcat.

4. View the cracked password `john --show aes_hash.txt`

Step 4: Cracking the DES-encrypted File with Hashcat

1. Install Hashcat (if not installed) `sudo apt install hashcat`

```
root@JackLinux:~# sudo apt install hashcat
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
hashcat is already the newest version (6.2.6+ds1-1build2).
0 upgraded, 0 newly installed, 0 to remove and 156 not upgraded.
root@JackLinux:~#
```

2. Run Hashcat with a Brute-Force Attack `hashcat -m 14000 des_hash.txt password_list.txt` 13

I couldn't get the original des encryption method to work because it was insecure. I made a new wordlist and hash to crack.

```
root@JackLinux:~# echo -n "password 123" | md5sum | awk '{print $1}' > test.hash
root@JackLinux:~# cat test.hash
6d2e1c6dd505a108cc7e19a46aa30a8a
```

```
root@JackLinux:~# cat > tinylist.txt <<'EOF'
> 123456
> password
> password123
> letmein
> qwerty
> EOF
root@JackLinux:~# cat tinylist.txt
123456
password
password123
letmein
qwerty
root@JackLinux:~#
```

```

Host memory required for this attack: 0 MB

Dictionary cache built:
* Filename...: tinylist.txt
* Passwords..: 5
* Bytes.....: 43
* Keyspace...: 5
* Runtime....: 0 secs

The wordlist or mask that you are using is too small.
This means that hashcat cannot use the full parallel power of your device(s).
Unless you supply more work, your cracking speed will drop.
For tips on supplying more work, see: https://hashcat.net/faq/morework

Approaching final keyspace - workload adjusted.

Session.....: hashcat
Status.....: Exhausted
Hash.Mode.....: 0 (MD5)
Hash.Target.....: 6d2e1c6dd505a108cc7e19a46aa30a8a
Time.Started.....: Thu Nov 13 21:54:45 2025 (0 secs)
Time.Estimated...: Thu Nov 13 21:54:45 2025 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (tinylist.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 59 H/s (0.00ms) @ Accel:1024 Loops:1 Thr:1 Vec:16
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 5/5 (100.00%)
Rejected.....: 0/5 (0.00%)
Restore.Point....: 5/5 (100.00%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: 123456 -> qwerty
Hardware.Mon.#1..: Temp: 64c Util: 98%

Started: Thu Nov 13 21:54:06 2025
Stopped: Thu Nov 13 21:54:46 2025
root@JackLinux:~#

```

3. Check if the password was cracked hashcat --show des_hash.txt

```

root@JackLinux:~# hashcat --show test.hash
The following 11 hash-modes match the structure of your input hash:

# | Name | Category
---+-----+-----
900 | MD4 | Raw Hash
0 | MD5 | Raw Hash
70 | md5(utf16le($pass)) | Raw Hash
2600 | md5(md5($pass)) | Raw Hash salted and/or iterated
3500 | md5(md5(md5($pass))) | Raw Hash salted and/or iterated
4400 | md5(sha1($pass)) | Raw Hash salted and/or iterated
20900 | md5(sha1($pass).md5($pass).sha1($pass)) | Raw Hash salted and/or iterated
4300 | md5(strtoupper(md5($pass))) | Raw Hash salted and/or iterated
1000 | NTLM | Operating System
9900 | Radmin2 | Operating System
8600 | Lotus Notes/Domino 5 | Enterprise Application Software (EAS

Please specify the hash-mode with -m [hash-mode].

root@JackLinux:~# hashcat --show -m test.hash
The specified parameter cannot use 'test.hash' as a value - must be a number.

root@JackLinux:~# cat test.hash
6d2e1c6dd505a108cc7e19a46aa30a8a

```